

Integral, Transform, and Fractional-Order Frontiers for the Fabius–Rvachev System

Weighted and repeated antiderivatives, inverse quantile calculus,
nonlocal fractional operators, resolvent hierarchies, and sinc-product energies

Research report prepared for the ProveIt Fabius-function program

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Abstract

This report develops a broad integral calculus for the Fabius function F , its inverse $Q = F^{-1}$, and Rvachev’s compactly supported function up, after a recursive audit of the current and archived LaTeX corpus under the ProveIt documentation root. Existing dyadic derivative formulas, exact moment arithmetic, q-binomial and Thue–Morse structure, sinc products, Lambert– W endpoint expansions, Bell and Bernoulli polynomial systems, repeated integration, weighted primitives, inverse-area identities, and the known obstruction to naive fractional self-similarity are treated as baseline results rather than reannounced as new.

The report first synthesizes ordinary and weighted antiderivatives, definite integrals, entire Mellin moments, negative and complex powers, Laplace/Fourier/Mellin/Stieltjes transforms, inverse-function dualities, order-statistic identities, and all integer orders of differentiation and integration. It then advances beyond the audited corpus in four directions. First, the inverse Fabius endpoint is placed in a fractional calculus: for $T = \log(1/y)$ and $\eta = yQ'(y)/Q(y) \sim \sqrt{\log 2/(2T)}$,

$$I_{0+}^a(Q^\beta)(y) = \frac{y^a Q(y)^\beta}{\Gamma(a+1)} (1 - \beta H_a \eta + o(\eta)),$$

while the Riemann–Liouville/Caputo derivative satisfies the dual digamma correction and the sharp strong threshold $D_{0+}^\sigma(Q^\beta) \in L^p \iff \sigma p \leq 1$ near the endpoint. Second, exact exterior moment series are obtained for the Riesz potential and fractional Laplacian of up; all coefficients are generated rationally from Bernoulli cumulants through complete Bell polynomials. Third, the Cauchy transform is shown to obey a logarithmic smoothing fixed point and an all-order resolvent hierarchy. Fourth, the log-periodic obstruction to fractional dyadic scaling is converted into an exact Floquet multiplier cocycle that restores the continuous dilation orbit on the global signed Fabius extension.

A deterministic, fully commented Python program reconstructs up from its infinite sinc product and checks the new nonlocal series, transform identities, and fractional endpoint corrections. “New” means not found in the recursively audited repository corpus at the time of writing; no claim of worldwide priority is made without a separate literature proof.

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1 Corpus audit, scope, and novelty boundary

1.1 What was reviewed

The repository was traversed recursively through the documentation root, including the main monographs, glossary, Lean walkthrough, non-elementarity article, papers, canonical research-frontier volume, current frontier drafts, auxiliary generated TeX fragments, and frozen archive. The archive map was used to identify historical duplicates and their canonical successors; living sources and current drafts were checked directly at theorem and section level, with integration-specific full-text searches described in Appendix C.

The audit is anchored to the path-complete inventory at commit 32d6d36c51d803289e6d6a0dc0c37753766eba47, which records exactly sixty-seven LaTeX paths. A second traversal of the live main branch found five later frontier packages adding eleven visible TeX paths, including the self-sampling report's six generated table fragments. Thus the current ledger contains seventy-eight paths: five core synthesis/reference sources, thirty-two frozen historical sources, thirty-four active frontier-report or auxiliary sources, and seven paper transcriptions [1, 2, 3].

The following table states the boundary used in this report.

Imported baseline	How it is used here
Probability interpretation of F , compact support and refinement equations for up, and the infinite sinc product	Normalization and the common probabilistic language for all integrals.
Exact dyadic values, rational moment recurrences, Bernoulli and q -binomial structure	Inputs to the moment, Mellin, Newton-series, and transform calculations; these arithmetic results are not reproved in full.
Lambert– W endpoint and inverse asymptotics, periodic phase corrections	Imported as the asymptotic input for the endpoint fractional calculus and as comparison data for the Floquet phase.
Dyadic self-sampling quadrature, finite-prefix inverse acceleration, Rvachev–Appell, Newton/exponent-sequence, and inverse-moment polynomial systems	Explicitly excluded from novelty claims; only consequences needed for integration and transform formulas are recalled.
Canonical integration and repeated-integration notebooks, inverse-function reports, and weighted primitive identities	Treated as prior art. Ordinary inverse-pair, moment, and transform formulas below form a self-contained synthesis; the strongest corpus-relative claims are restricted to the fractional endpoint, nonlocal, resolvent, and Floquet results.

Strongest corpus-relative frontier results

After comparison with all seventy-eight paths, the following proved package was not located explicitly in the audited corpus: the Riemann–Liouville inverse-pair duality of [theorem 7.1](#); the digamma/trigamma endpoint expansions and sharp critical strong- L^p law for Q^β in [theorems 9.2](#) and [9.4](#); the exact exterior fractional-Laplacian and Riesz-potential series and the fractional Sobolev-energy identity in [theorems 10.1](#), [10.2](#) and [10.4](#); the logarithmic Cauchy-transform fixed point and all-order resolvent hierarchy of [theorem 12.1](#); and the Floquet multiplier/cocycle that repairs fractional dyadic scaling in [theorem 11.1](#). The sinc-product Gini identity of [theorem 13.2](#) is another compact deduction not found in the comparison searches. The universal inverse-pair identity, entire moment/Newton calculus, negative-moment recurrence, beta transforms, and related ordinary formulas are presented as a useful self-contained synthesis and auxiliary deductions, without a priority claim.

1.2 Conventions

All powers of positive real variables use the real logarithm: $x^s = \exp(s \log x)$. The totalized Fabius distribution function is

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1).$$

Its inverse $Q : (0, 1) \rightarrow (0, 1)$ is the ordinary increasing inverse; endpoint values are understood by continuity. Angular frequency is used for Fourier transforms: $\widehat{f}(t) = \int_{\mathbb{R}} e^{itx} f(x) dx$.

2 Probability model and transform baseline

2.1 The dyadic uniform sum

Let $U_1, U_2, \dots$ be independent random variables uniformly distributed on $[0, 1]$, and put

$$Y = \sum_{k=1}^{\infty} \frac{U_k}{2^k}, \quad Z = 2Y - 1 = \sum_{k=1}^{\infty} \frac{2U_k - 1}{2^k}. \quad (1)$$

Then $Y \in [0, 1]$, $Z \in [-1, 1]$, and the laws are symmetric:

$$Y \stackrel{d}{=} 1 - Y, \quad Z \stackrel{d}{=} -Z. \quad (2)$$

The Fabius function is the distribution function of Y ,

$$F(x) = \mathbb{P}(Y \leq x), \quad (3)$$

and up is the density of Z . Thus

$$F'(x) = 2 \text{up}(2x - 1), \quad \text{up}(x) = \begin{cases} F(x + 1), & -1 \leq x \leq 0, \\ F(1 - x) = 1 - F(x), & 0 \leq x \leq 1, \\ 0, & |x| > 1. \end{cases} \quad (4)$$

In particular $F(x) + F(1 - x) = 1$ on $[0, 1]$, and the quantile obeys

$$Q(1 - v) = 1 - Q(v). \quad (5)$$

The self-decomposition

$$Y \stackrel{d}{=} \frac{U + Y'}{2} \quad (6)$$

holds with $U \sim \text{Unif}[0, 1]$ independent of a copy Y' of Y . It will generate both the positive- and negative-moment recurrences.

2.2 Moment-generating and Fourier products

Define

$$P(t) = \mathbb{E}e^{tY}, \quad \mathcal{U}(t) = \mathbb{E}e^{tZ}. \quad (7)$$

Independence in [equation \(1\)](#) gives the locally uniformly convergent products

$$P(t) = \prod_{k=1}^{\infty} \frac{e^{t/2^k} - 1}{t/2^k}, \quad (8)$$

$$\mathcal{U}(t) = \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k}, \quad (9)$$

$$P(t) = e^{t/2} \mathcal{U}(t/2). \quad (10)$$

The characteristic function of the up law is therefore

$$\phi(t) = \widehat{\text{up}}(t) = \prod_{k=1}^{\infty} \frac{\sin(t/2^k)}{t/2^k}. \quad (11)$$

The scaling equation

$$P(2t) = \frac{e^t - 1}{t} P(t) \quad (12)$$

is the exponential generating-function form of the moment recurrence below.

2.3 Rational moments, cumulants, and Bell polynomials

Write

$$m_n = \mathbb{E}Y^n = \int_0^1 Q(v)^n dv, \quad \mu_n = \mathbb{E}Z^n = \int_{-1}^1 x^n \text{up}(x) dx. \quad (13)$$

Taking the n -th moment in [equation \(6\)](#) gives

$$(2^n - 1)m_n = \sum_{k=0}^{n-1} \binom{n}{k} \frac{m_k}{n - k + 1}, \quad m_0 = 1. \quad (14)$$

Thus every m_n is rational. The first values are

$$1, \frac{1}{2}, \frac{5}{18}, \frac{1}{6}, \frac{143}{1350}, \frac{19}{270}, \frac{1153}{23814}, \frac{583}{17010}, \dots \quad (15)$$

The logarithm of [equation \(9\)](#) and $\log(\sinh z/z) = \sum_{r \geq 1} 2^{2r-1} B_{2r} z^{2r} / (r(2r)!)$ give the cumulants

$$\kappa_1(Y) = \frac{1}{2}, \quad \kappa_{2r}(Y) = \frac{B_{2r}}{2r(2^{2r} - 1)}, \quad \kappa_{2r+1}(Y) = 0 \quad (r \geq 1), \quad (16)$$

$$\kappa_{2r}(Z) = \frac{2^{2r} B_{2r}}{2r(2^{2r} - 1)}, \quad \kappa_{2r+1}(Z) = 0. \quad (17)$$

Consequently m_n and μ_n are complete exponential Bell polynomials in these cumulants. A useful bridge between the one-sided and centered moments is

$$\mu_{2j} = 2 \int_0^1 x^{2j} \text{up}(x) dx = \frac{2m_{2j+1}}{2j+1}, \quad \mu_{2j+1} = 0. \quad (18)$$

For example

$$\mu_2 = \frac{1}{9}, \quad \mu_4 = \frac{19}{675}, \quad \mu_6 = \frac{583}{59535}. \quad (19)$$

3 A universal integration calculus for F and Q

The classical area identity for a function and its inverse is only the first member of a much larger family. The following theorem is the main algebraic engine of the report.

Theorem 3.1 (two-function inverse-pair identity). *Let A, B be absolutely continuous on $[0, 1]$. For every $z \in [0, 1]$,*

$$A(z)B(F(z)) - A(0)B(0) = \int_0^z A'(x)B(F(x)) dx + \int_0^{F(z)} A(Q(v))B'(v) dv. \quad (20)$$

At $z = 1$,

$$A(1)B(1) - A(0)B(0) = \int_0^1 A'(x)B(F(x)) dx + \int_0^1 A(Q(v))B'(v) dv. \quad (21)$$

The formulas remain valid under the usual improper-integral hypotheses whenever an endpoint is singular.

Proof. Differentiate the product $A(x)B(F(x))$, integrate from 0 to z , and in the term containing $F'(x)$ use the increasing substitution $v = F(x)$, so that $x = Q(v)$. $\square$

Remark 3.2. No special functional equation of the Fabius function is needed in [theorem 3.1](#); it is a quantile identity. What is special here is that the moments and transforms of this particular quantile can be calculated by dyadic products and rational recurrences.

3.1 Power duality and logarithmic derivatives

Taking $A(x) = x^p$, $B(v) = v^q$ gives the following Mellin-type duality.

Corollary 3.3 (power duality). *Initially for $\Re p, \Re q > 0$,*

$$p \int_0^z x^{p-1} F(x)^q dx + q \int_0^{F(z)} Q(v)^p v^{q-1} dv = z^p F(z)^q. \quad (22)$$

In particular,

$$p \int_0^1 x^{p-1} F(x)^q dx + q \int_0^1 v^{q-1} Q(v)^p dv = 1. \quad (23)$$

Both sides admit meromorphic or entire continuation in parameters whenever the endpoint flatness cancels the apparent singularities.

For all logarithmic derivatives at once, define

$$J_{r,s}(p, q) = \int_0^1 x^{p-1} (\log x)^r F(x)^q (\log F(x))^s dx, \quad (24)$$

$$K_{r,s}(p, q) = \int_0^1 v^{q-1} (\log v)^s Q(v)^p (\log Q(v))^r dv. \quad (25)$$

Differentiating [equation \(23\)](#) gives, for $r + s > 0$,

$$pJ_{r,s} + rJ_{r-1,s} + qK_{r,s} + sK_{r,s-1} = 0, \quad (26)$$

with $pJ_{0,0} + qK_{0,0} = 1$. Equation [equation \(26\)](#) is a compact general formula for every mixed derivative of the two Mellin kernels.

3.2 A bivariate Laplace–Fourier identity

The exponential choice $A(x) = e^{sx}$, $B(v) = e^{tv}$ produces an especially simple transform relation.

Theorem 3.4 (bivariate inverse-pair transform). *For all $s, t \in \mathbb{C}$ and $z \in [0, 1]$,*

$$s \int_0^z e^{sx+tF(x)} dx + t \int_0^{F(z)} e^{tv+sQ(v)} dv = e^{sz+tF(z)} - 1. \quad (27)$$

Hence

$$s \int_0^1 e^{sx+tF(x)} dx + t \int_0^1 e^{tv+sQ(v)} dv = e^{s+t} - 1. \quad (28)$$

The identities are entire in (s, t) ; the cases $s = 0$ or $t = 0$ are understood by continuity.

Every mixed power integral involving x and $F(x)$, together with its quantile dual, is obtained by coefficient extraction from [equation \(28\)](#). Purely imaginary s, t give a bivariate Fourier identity.

3.3 Weighted antiderivatives

Taking $B(v) = v$ and a primitive A of a prescribed weight yields the useful master formula

$$\int_0^x A'(t) F(t) dt = A(x) F(x) - A(0) F(0) - \int_0^{F(x)} A(Q(v)) dv. \quad (29)$$

For monomial weights this becomes the promised indefinite integral.

Corollary 3.5 (monomial–Fabius antiderivative). *For $p \neq -1$, initially where the integrals converge and then by continuation,*

$$\int_0^x t^p F(t) dt = \frac{x^{p+1} F(x) - \int_0^{F(x)} Q(v)^{p+1} dv}{p+1}. \quad (30)$$

The value at $p = -1$ is removable:

$$\int_0^x \frac{F(t)}{t} dt = F(x) \log x - \int_0^{F(x)} \log Q(v) dv. \quad (31)$$

On $[0, 1]$, where $\text{up} = 1 - F$,

$$\int_0^x t^p \text{up}(t) dt = \frac{x^{p+1} (1 - F(x)) + \int_0^{F(x)} Q(v)^{p+1} dv}{p+1}. \quad (32)$$

More generally, from [equation \(22\)](#),

$$\int_0^x t^{p-1} F(t)^q dt = \frac{x^p F(x)^q}{p} - \frac{q}{p} \int_0^{F(x)} Q(v)^p v^{q-1} dv. \quad (33)$$

Thus products of F and the positive half of up reduce to beta-weighted quantile integrals after expanding $(1 - F)^n$.

4 The entire complex moment function

Definition 4.1. For $s \in \mathbb{C}$, set

$$\mathfrak{M}(s) = \mathbb{E}Y^s = \int_0^1 Q(v)^s dv. \quad (34)$$

For nonnegative integers, $\mathfrak{M}(n) = m_n$.

4.1 Entire continuation

The endpoint flatness of F is much stronger than what is needed for ordinary positive moments: for every $N > 0$,

$$F(x) = O_N(x^N) \quad (x \downarrow 0). \quad (35)$$

Theorem 4.2 (entire quantile Mellin function). *The function $\mathfrak{M}$ in [equation \(34\)](#) is entire. For every $r \geq 0$,*

$$\mathfrak{M}^{(r)}(s) = \int_0^1 Q(v)^s (\log Q(v))^r dv = \mathbb{E}[Y^s (\log Y)^r]. \quad (36)$$

In particular all negative moments and all logarithmic moments exist.

Proof. Let $K \subset \mathbb{C}$ be compact and choose $A > 0$ with $\Re s \geq -A$ on K . For every fixed r , the integrand is bounded in modulus by $Y^{-A} |\log Y|^r$. Integration by parts gives

$$\mathbb{E}Y^{-A} = 1 + A \int_0^1 x^{-A-1} F(x) dx < \infty,$$

using [equation \(35\)](#) with an exponent larger than A ; inserting powers of $|\log x|$ does not alter convergence. Dominated differentiation on compact sets proves holomorphy and [equation \(36\)](#). $\square$

A first transform triangle follows immediately:

$$\int_0^1 x^{s-1} F(x) dx = \frac{1 - \mathfrak{M}(s)}{s}, \quad (37)$$

$$\int_0^1 x^{s-1} \text{up}(x) dx = \frac{\mathfrak{M}(s)}{s}, \quad (38)$$

$$\int_0^1 Q(v)^{s-1} dv = \mathfrak{M}(s-1). \quad (39)$$

The first quotient has a removable singularity at $s = 0$ and is entire; the second is the meromorphic continuation of the classical Mellin transform of $\text{up}|_{[0,1]}$ and has exactly one pole, at $s = 0$, with residue 1. Evenness gives

$$\int_{-1}^1 |x|^{s-1} \text{up}(x) dx = \frac{2\mathfrak{M}(s)}{s}. \quad (40)$$

4.2 Superpolynomial moment decay

Lemma 4.3. *For every $N > 0$,*

$$m_k = O_N(k^{-N}) \quad (k \rightarrow \infty). \quad (41)$$

Proof. For $k > 0$, the tail formula and symmetry give

$$m_k = k \int_0^1 x^{k-1} (1 - F(x)) dx = k \int_0^1 x^{k-1} F(1-x) dx.$$

By [equation \(35\)](#), $F(1-x) \leq C_N(1-x)^N$, hence

$$m_k \leq C_N k B(k, N+1) = O(k^{-N}).$$

□

4.3 Two globally convergent Newton series

Theorem 4.4 (Newton and central-even series). *Both of the following series converge locally uniformly for all $s \in \mathbb{C}$:*

$$\mathfrak{M}(s) = \sum_{k=0}^{\infty} (-1)^k \binom{s}{k} m_k, \quad (42)$$

$$\mathfrak{M}(s) = 2^{-s} \sum_{j=0}^{\infty} \binom{s}{2j} \mu_{2j} = 2^{-s} \sum_{j=0}^{\infty} \binom{s}{2j} \frac{2m_{2j+1}}{2j+1}. \quad (43)$$

They therefore represent the entire function $\mathfrak{M}$ on the whole plane.

Proof. For $0 < Y < 1$, expand $Y^s = (1 - (1 - Y))^s$. Taking expectations and using $1 - Y \stackrel{d}{=} Y$ gives [equation \(42\)](#). On every compact set in s , the binomial coefficients have at most polynomial growth in k ; [theorem 4.3](#) therefore justifies absolute, locally uniform convergence and interchange of sum and expectation.

For [equation \(43\)](#), write $Y = (1 + Z)/2$, expand $(1 + Z)^s$, and use the evenness of the law of Z . The relation [equation \(18\)](#) and [theorem 4.3](#) again give locally uniform convergence. □

Several useful specializations are immediate. For an integer $r \geq 1$,

$$\mathfrak{M}(-r) = \sum_{k=0}^{\infty} \binom{r+k-1}{k} m_k. \quad (44)$$

Every term is rational and positive. In particular,

$$\mathfrak{M}(-1) = \sum_{k=0}^{\infty} m_k. \quad (45)$$

Differentiation of [equation \(42\)](#) at the origin yields logarithmic moments in terms of signed Stirling numbers of the first kind $s(k, r)$:

$$\mathfrak{M}^{(r)}(0) = r! \sum_{k=r}^{\infty} \frac{(-1)^k s(k, r)}{k!} m_k. \quad (46)$$

For $r = 1$,

$$-\mathbb{E} \log Y = \sum_{k=1}^{\infty} \frac{m_k}{k} = \int_0^1 \frac{F(x)}{x} dx. \quad (47)$$

Numerically,

$$-\mathbb{E} \log Y = 0.758240000404408656 \dots \quad (48)$$

4.4 A mixed reflected Mellin function

The inverse symmetry [equation \(5\)](#) suggests the two-variable transform

$$\mathfrak{M}_2(a, b) = \int_0^1 Q(v)^a Q(1-v)^b dv = \mathbb{E}[Y^a (1-Y)^b]. \quad (49)$$

Proposition 4.5. *The function $\mathfrak{M}_2$ is entire on $\mathbb{C}^2$, symmetric in a, b , and has the locally uniformly convergent expansion*

$$\mathfrak{M}_2(a, b) = \sum_{k=0}^{\infty} (-1)^k \binom{b}{k} \mathfrak{M}(a+k). \quad (50)$$

It also has the central expansion

$$\mathfrak{M}_2(a, b) = 2^{-a-b} \sum_{j,k \geq 0} (-1)^k \binom{a}{j} \binom{b}{k} \mu_{j+k}. \quad (51)$$

If either parameter is a nonnegative integer, the corresponding binomial expansion terminates.

Proof. Endpoint flatness at both 0 and 1 gives local domination for arbitrary negative real parts, proving entire dependence on two parameters. Expand $(1-Y)^b$ to obtain [equation \(50\)](#); superpolynomial decay of shifted moments gives local uniformity. The central formula follows from $Y = (1+Z)/2$, $1-Y = (1-Z)/2$. $\square$

For example,

$$\int_0^1 Q(v) Q(1-v) dv = \frac{2}{9}, \quad \int_0^1 Q(v)^2 Q(1-v)^2 dv = \frac{34}{675}, \quad (52)$$

and

$$\int_0^1 \frac{dv}{Q(v)(1-Q(v))} = 2\mathfrak{M}(-1). \quad (53)$$

5 Negative moments: an exact recurrence and a new asymptotic constant

The Newton series [equation \(44\)](#) is effective for moderate negative orders. A second construction is much more stable for large orders and reveals a new normalization law.

5.1 The recurrence

For $s > 0$, define

$$A(s) = \mathfrak{M}(-s) = \mathbb{E}Y^{-s}, \quad D(s) = \mathbb{E}(1 + Y)^{-s}. \quad (54)$$

Conditioning in [equation \(6\)](#) and integrating the uniform variable gives the following exact relation.

Theorem 5.1 (negative-moment recurrence). *For every real $s > 0$,*

$$A(s + 1) = \frac{2^{s+1}}{s} (A(s) - D(s)). \quad (55)$$

Equivalently, for $s > 1$,

$$A(s) = \frac{2^s}{s-1} (A(s-1) - D(s-1)). \quad (56)$$

Moreover $D(s)$ has the rapidly convergent centered series

$$D(s) = \left(\frac{2}{3}\right)^s \sum_{j=0}^{\infty} \frac{(s)_{2j}}{(2j)! 3^{2j}} \mu_{2j}, \quad (57)$$

where $(s)_n = s(s+1) \cdots (s+n-1)$.

Proof. By [equation \(6\)](#), for $s > 0$,

$$A(s + 1) = 2^{s+1} \mathbb{E} \int_0^1 (U + Y')^{-s-1} dU.$$

For $y > 0$,

$$\int_0^1 (u + y)^{-s-1} du = \frac{y^{-s} - (1 + y)^{-s}}{s},$$

which gives [equation \(55\)](#). For [equation \(57\)](#), write $1 + Y = (3 + Z)/2$, expand $(1 + Z/3)^{-s}$, and use the vanishing of odd moments. The factor 3^{-2j} makes the series geometrically stable in addition to the superpolynomial decay of μ_{2j} . $\square$

The first values generated by this recurrence are

$$\begin{aligned} A(1) &= 2.31488614400581905 \dots, \\ A(2) &= 6.55899183845822858 \dots, \\ A(3) &= 24.3890791927221220 \dots, \\ A(4) &= 128.368907270620532 \dots, \\ A(5) &= 1025.15473735960333 \dots \end{aligned} \quad (58)$$

5.2 A normalized limit and a periodic phase profile

Define

$$H(s) = \frac{\Gamma(s)A(s)}{2^{s(s+1)/2}}, \quad \rho(s) = \frac{D(s)}{A(s)}. \quad (59)$$

Then [equation \(55\)](#) becomes

$$H(s+1) = H(s)(1 - \rho(s)). \quad (60)$$

Both factors are positive. Since $\mathbb{P}(Y \leq 1/2) = 1/2$,

$$A(s) \geq 2^{s-1}, \quad D(s) \leq 1, \quad 0 < \rho(s) \leq 2^{1-s}. \quad (61)$$

This elementary estimate is enough to prove convergence.

Theorem 5.2 (periodic limiting profile). *For $0 < \theta \leq 1$, the product*

$$C(\theta) = H(\theta) \prod_{j=0}^{\infty} (1 - \rho(\theta + j)) \quad (62)$$

converges to a positive number. It extends by $C(\theta + 1) = C(\theta)$ to a positive, continuous, one-periodic function. As $n \rightarrow \infty$,

$$A(n + \theta) = C(\theta) \frac{2^{(n+\theta)(n+\theta+1)/2}}{\Gamma(n + \theta)} (1 + O_{\theta}(2^{-n})). \quad (63)$$

For integer $r \rightarrow \infty$,

$$\boxed{\mathfrak{M}(-r) = C_* \frac{2^{r(r+1)/2}}{\Gamma(r)} (1 + O(2^{-r}))}, \quad C_* = C(1) > 0. \quad (64)$$

Proof. Iteration of [equation \(60\)](#) gives

$$H(n + \theta) = H(\theta) \prod_{j=0}^{n-1} (1 - \rho(\theta + j)).$$

The sum of the tail bounds in [equation \(61\)](#) is geometric. Hence the infinite product converges to a positive limit, and its omitted tail differs from 1 by $O_{\theta}(2^{-n})$. This proves [equation \(63\)](#). Shifting the initial point by one and using [equation \(60\)](#) proves exact periodicity. Uniform convergence on compact subintervals gives continuity; exact periodicity then removes the apparent endpoint at $\theta = 0$. The integer formula is the case $\theta = 1$. $\square$

For integers the normalized sequence

$$C_r = \frac{(r-1)! \mathfrak{M}(-r)}{2^{r(r+1)/2}} \quad (65)$$

is strictly decreasing and satisfies

$$C_{r+1} = C_r \left(1 - \frac{D(r)}{A(r)}\right), \quad C_r \downarrow C_*. \quad (66)$$

The product representation is

$$C_* = \frac{\mathfrak{M}(-1)}{2} \prod_{j=1}^{\infty} \left(1 - \frac{D(j)}{A(j)}\right). \quad (67)$$

The computation described in [section 15](#) gives

$$C_* = 0.750721980407870959 \dots \quad (68)$$

The rapid stabilization is visible in

r	1	2	4	6	8	12
C_r	1.15744307	0.81987398	0.75216157	0.75072879	0.75072199	0.7507219804

5.3 Logarithmic asymptotics and entire-function growth

Stirling's expansion in [equation \(64\)](#) gives a complete algebraic asymptotic series:

$$\begin{aligned} \log \mathfrak{M}(-r) = & \frac{\log 2}{2} r^2 - r \log r + \left(1 + \frac{\log 2}{2}\right) r + \frac{1}{2} \log r + \log \frac{C_*}{\sqrt{2\pi}} \\ & - \frac{1}{12r} + \frac{1}{360r^3} - \frac{1}{1260r^5} + \cdots, \end{aligned} \quad (69)$$

where

$$\log \frac{C_*}{\sqrt{2\pi}} = -1.20565842815389633 \dots \quad (70)$$

This refines the growth one would obtain by a direct saddle-point transfer from the Lambert– W endpoint expansion.

There is also a clean entire-function consequence. If $M_{\mathfrak{M}}(R) = \max_{|s|=R} |\mathfrak{M}(s)|$, then

$$|\mathfrak{M}(s)| \leq \mathbb{E} Y^{\Re s} \leq \mathbb{E} Y^{-R} = \mathfrak{M}(-R), \quad |s| = R, \quad (71)$$

while equality occurs at $s = -R$. Thus

$$M_{\mathfrak{M}}(R) = \mathfrak{M}(-R). \quad (72)$$

Corollary 5.3. *The entire function $\mathfrak{M}$ has order 2 and type $(\log 2)/2$.*

5.4 Singular Fabius integrals

Putting $s = p + 1$ in [equation \(37\)](#) shows that

$$\int_0^1 x^p F(x) dx = \frac{1 - \mathfrak{M}(p + 1)}{p + 1} \quad (73)$$

is an entire function of p , despite the apparent pole at $p = -1$. Therefore all negative powers are integrable against F . For integers $k \geq 2$,

$$\int_0^1 \frac{F(x)}{x^k} dx = \frac{\mathfrak{M}(1 - k) - 1}{k - 1}. \quad (74)$$

The first examples are

$$\begin{aligned} \int_0^1 \frac{F(x)}{x} dx &= 0.758240000404408656 \dots, \\ \int_0^1 \frac{F(x)}{x^2} dx &= 1.31488614400581905 \dots, \\ \int_0^1 \frac{F(x)}{x^3} dx &= 2.77949591922911429 \dots, \\ \int_0^1 \frac{F(x)}{x^4} dx &= 7.79635973090737400 \dots \end{aligned} \quad (75)$$

By [equation \(64\)](#),

$$\int_0^1 \frac{F(x)}{x^k} dx \sim C_* \frac{2^{k(k-1)/2}}{\Gamma(k)}. \quad (76)$$

This is an asymptotic law for a natural family of increasingly singular antiderivatives.

6 Definite weighted integrals

6.1 Polynomial and analytic weights

Let g be integrable and let $G(x) = \int_0^x g(t) dt$. Taking $A = G$, $B(v) = v$ in [theorem 3.1](#) gives

$$\int_0^1 g(x)F(x) dx = G(1) - \mathbb{E}G(Y), \quad (77)$$

$$\int_0^1 g(x)(1 - F(x)) dx = \mathbb{E}G(Y). \quad (78)$$

For a polynomial g , every quantity on the right is a rational combination of the moments m_n . For an entire g , the same formulas give convergent series whenever the Taylor expansion can be integrated termwise.

The monomial cases are

$$\int_0^1 x^p F(x) dx = \frac{1 - \mathfrak{M}(p+1)}{p+1}, \quad (79)$$

$$\int_0^1 x^p \text{up}(x) dx = \frac{\mathfrak{M}(p+1)}{p+1} \quad (\text{up} = 1 - F \text{ on } [0, 1]). \quad (80)$$

For $p \in \mathbb{N}$ these are rational. For example,

p	0	1	2	3
$\int_0^1 x^p F(x) dx$	1/2	13/36	5/18	1207/5400
$\int_0^1 x^p \text{up}(x) dx$	1/2	5/36	1/18	143/5400

For symmetric weights $g(x) = g(1 - x)$, symmetry alone gives

$$\int_0^1 g(x)F(x) dx = \frac{1}{2} \int_0^1 g(x) dx. \quad (81)$$

6.2 Bernoulli-polynomial weights

Let

$$\beta_n = \mathbb{E}B_n(Y). \quad (82)$$

Their exponential generating function follows from [equation \(8\)](#):

$$\sum_{n=0}^{\infty} \beta_n \frac{t^n}{n!} = \frac{tP(t)}{e^t - 1}. \quad (83)$$

Symmetry implies $\beta_{2j+1} = 0$. Since $B'_{n+1} = (n+1)B_n$, [equation \(77\)](#) gives, for $n \geq 1$,

$$\int_0^1 B_n(x)F(x) dx = -\frac{\beta_{n+1} - B_{n+1}}{n+1}. \quad (84)$$

The first nontrivial expected Bernoulli values are

$$\beta_2 = -\frac{1}{18}, \quad \beta_4 = \frac{23}{1350}, \quad \beta_6 = -\frac{1543}{119070}, \quad \beta_8 = \frac{1196113}{65063250}. \quad (85)$$

This supplies a direct integral bridge between the Bernoulli-polynomial structure and the Fabius distribution.

6.3 Beta weights

For $\Re a, \Re b > 0$, define

$$I_F(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} F(x) dx. \quad (86)$$

Theorem 6.1 (beta-weighted Fabius integrals). *For $\Re a, \Re b > 0$,*

$$I_F(a, b) = \mathbb{E} B_Y(b, a) = \sum_{n=0}^{\infty} \frac{(1-a)_n}{n!(b+n)} \mathfrak{M}(b+n), \quad (87)$$

$$\int_0^1 x^{a-1} (1-x)^{b-1} (1-F(x)) dx = \mathbb{E} B_Y(a, b) = \sum_{n=0}^{\infty} \frac{(1-b)_n}{n!(a+n)} \mathfrak{M}(a+n). \quad (88)$$

Here B_y is the incomplete beta function. If a in [equation \(87\)](#), or b in [equation \(88\)](#), is a positive integer, the corresponding series terminates. If both parameters are positive integers, the integrals are rational.

Proof. Use $F(x) = \mathbb{E} \mathbf{1}_{\{Y \leq x\}}$ and Fubini:

$$I_F(a, b) = \mathbb{E} \int_Y^1 x^{a-1} (1-x)^{b-1} dx = \mathbb{E} B_{1-Y}(b, a) = \mathbb{E} B_Y(b, a),$$

where the last equality uses $1-Y \stackrel{d}{=} Y$. The hypergeometric expansion

$$B_y(b, a) = \sum_{n=0}^{\infty} \frac{(1-a)_n}{n!(b+n)} y^{b+n}$$

gives [equation \(87\)](#). The endpoint flatness and [theorem 4.3](#) justify convergence beyond the ordinary boundary restrictions of the pointwise hypergeometric series. The second identity is analogous. $\square$

For example, symmetry or direct termination gives

$$\int_0^1 x(1-x)F(x) dx = \frac{1}{12}. \quad (89)$$

Differentiating [equation \(87\)](#) in a and b produces all beta-logarithmic integrals with powers of $\log x$ and $\log(1-x)$.

6.4 A dyadic half-interval identity

The lower-half differential equation $F'(x) = 2F(2x)$ for $0 \leq x \leq 1/2$ leads to exact quantile truncations.

Proposition 6.2. *For every complex p , with removable values interpreted by continuation,*

$$\int_0^{1/2} Q(v)^p dv = 2^{-p} \frac{1 - \mathfrak{M}(p+1)}{p+1}, \quad (90)$$

$$\int_0^{1/2} x^p F(x) dx = \frac{2^{-p-1}}{p+1} \left[\frac{1}{2} - \frac{1 - \mathfrak{M}(p+2)}{p+2} \right]. \quad (91)$$

In particular,

$$\int_0^{1/2} F(x) dx = \frac{5}{72}. \quad (92)$$

Proof. Change variables $v = F(x)$ in the first integral and use the lower-half equation:

$$\int_0^{1/2} Q(v)^p dv = \int_0^{1/2} x^p F'(x) dx = 2^{-p} \int_0^1 y^p F(y) dy.$$

This is [equation \(90\)](#). Substitute it into the incomplete formula [equation \(30\)](#) at $x = 1/2$ to obtain [equation \(91\)](#). $\square$

7 Fractional and integer antiderivatives

Repeated integration has a particularly transparent probabilistic meaning. The same calculus extends without change from integer order to the Riemann–Liouville operator. For $\Re \alpha > 0$, write

$$(I_0^\alpha f)(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t) dt. \quad (93)$$

7.1 Fractional inverse-pair duality

Theorem 7.1 (fractional power duality). *Let $\Re \alpha > 0$. Initially for $\Re q > 0$,*

$$(I_0^\alpha F^q)(x) = \frac{q}{\Gamma(\alpha+1)} \int_0^{F(x)} (x-Q(v))^\alpha v^{q-1} dv. \quad (94)$$

Initially for $\Re p > 0$,

$$(I_0^\alpha Q^p)(v) = \frac{p}{\Gamma(\alpha+1)} \int_0^{Q(v)} (v-F(x))^\alpha x^{p-1} dx. \quad (95)$$

Both formulas continue in p or q wherever the endpoint terms admit continuation.

Proof. Since $d(x-t)^\alpha = -\alpha(x-t)^{\alpha-1} dt$, integration by parts gives

$$I_0^\alpha F^q(x) = \frac{1}{\Gamma(\alpha+1)} \int_0^x (x-t)^\alpha d(F(t)^q).$$

The boundary terms vanish when $\Re q > 0$. Using $d(F^q) = qF^{q-1} dF$ and substituting $v = F(t)$ proves [equation \(94\)](#). The proof of [equation \(95\)](#) is identical after interchanging the inverse pair. $\square$

The case $q = 1$ has a direct truncated-moment interpretation.

Corollary 7.2. *For $\Re \alpha > 0$,*

$$(I_0^\alpha F)(x) = \frac{\mathbb{E}[(x-Y)_+^\alpha]}{\Gamma(\alpha+1)}. \quad (96)$$

In particular,

$$(I_0^\alpha F)(1) = \frac{\mathfrak{M}(\alpha)}{\Gamma(\alpha+1)}. \quad (97)$$

Proof. In [equation \(94\)](#), the integral over v is the expectation of $(x-Q(V))_+^\alpha$ for a uniform V . Since $Q(V) \stackrel{d}{=} Y$, this is [equation \(96\)](#). At $x = 1$, use $1-Y \stackrel{d}{=} Y$. $\square$

Formula [equation \(97\)](#) says that the entire complex moment function is exactly the endpoint value of the fractional antiderivative, multiplied by $\Gamma(\alpha+1)$. Thus the Newton series in [equation \(42\)](#) is at the same time a global series for fractional repeated integration.

7.2 Repeated primitives of Rvachev's up function

For $\Re \alpha > 0$, define the left-anchored fractional integral

$$(I_{-1}^\alpha \text{up})(x) = \frac{1}{\Gamma(\alpha)} \int_{-1}^x (x-t)^{\alpha-1} \text{up}(t) dt. \quad (98)$$

Because up is the density of Z ,

$$(I_{-1}^\alpha \text{up})(x) = \frac{\mathbb{E}[(x-Z)_+^{\alpha-1}]}{\Gamma(\alpha)}. \quad (99)$$

At the right endpoint,

$$(I_{-1}^{\alpha} \text{up})(1) = \frac{2^{\alpha-1} \mathfrak{M}(\alpha-1)}{\Gamma(\alpha)}. \quad (100)$$

For an integer $n \geq 1$, [equation \(99\)](#) becomes

$$(I_{-1}^n \text{up})(x) = \frac{1}{(n-1)!} \mathbb{E}[(x-Z)_+^{n-1}]. \quad (101)$$

7.3 Appell polynomial tails

For $x \geq 1$, the positive-part signs in [equations \(96\)](#) and [\(101\)](#) disappear. Hence

$$(I_0^n F)(x) = \frac{1}{n!} \mathbb{E}(x-Y)^n, \quad x \geq 1, \quad (102)$$

$$(I_{-1}^n \text{up})(x) = \frac{1}{(n-1)!} \mathbb{E}(x-Z)^{n-1}, \quad x \geq 1. \quad (103)$$

The polynomial sequences

$$P_n(x) = \mathbb{E}(x-Y)^n, \quad R_n(x) = \mathbb{E}(x-Z)^n \quad (104)$$

are Appell sequences:

$$P'_n(x) = nP_{n-1}(x), \quad R'_n(x) = nR_{n-1}(x). \quad (105)$$

Their exponential generating functions are

$$\sum_{n \geq 0} P_n(x) \frac{t^n}{n!} = e^{tx} P(-t), \quad (106)$$

$$\sum_{n \geq 0} R_n(x) \frac{t^n}{n!} = e^{tx} \mathcal{U}(-t). \quad (107)$$

These polynomial tails and their inverse-moment projection role are part of the existing Rvachev–Appell research in the repository; here they arise as the exterior part of the fractional-truncated-moment formula.

7.4 Ordinary n -fold antiderivatives as incomplete moments

For an integer $n \geq 1$, the anchored n -fold antiderivative of F is

$$I_0^n F(x) = \frac{1}{n!} \int_0^{F(x)} (x-Q(v))^n dv. \quad (108)$$

The same expression can be expanded finitely:

$$I_0^n F(x) = \frac{1}{n!} \sum_{j=0}^n (-1)^j \binom{n}{j} x^{n-j} \int_0^{F(x)} Q(v)^j dv. \quad (109)$$

Thus every repeated primitive is determined by incomplete quantile moments. At a dyadic x , these can in turn be reduced by the exact dyadic machinery already developed in the repository.

For inverse powers, [equation \(95\)](#) gives the complementary expression

$$I_0^n (Q^p)(v) = \frac{p}{n!} \int_0^{Q(v)} x^{p-1} (v-F(x))^n dx. \quad (110)$$

The special case $p = 1$ is a fractional extension of the area identity:

$$I_0^n Q(v) = \frac{1}{n!} \int_0^{Q(v)} (v - F(x))^n dx. \quad (111)$$

For $n = 1$, this is

$$\int_0^v Q(w) dw = \int_0^{Q(v)} (v - F(x)) dx, \quad (112)$$

the usual area formula in a form adapted to the Fabius quantile.

8 Derivatives of arbitrary order

8.1 Derivatives of the Fabius function

On the lower dyadic branch, repeated differentiation of $F'(x) = 2F(2x)$ gives

$$F^{(n)}(x) = 2^{n(n+1)/2} F(2^n x), \quad 0 < x < 2^{-n}. \quad (113)$$

The full interval is obtained by the known dyadic/Thue–Morse reflection formulas. We do not duplicate those formulas here; the role of [equation \(113\)](#) is to show why every inverse derivative becomes extremely singular near the endpoints.

8.2 Inverse derivatives and Bell polynomials

Let $f_j = F^{(j)}(x)$. Define rational differential expressions recursively by

$$\mathcal{R}_1(x) = \frac{1}{F'(x)}, \quad \mathcal{R}_{n+1}(x) = \frac{1}{F'(x)} \frac{d}{dx} \mathcal{R}_n(x). \quad (114)$$

Proposition 8.1 (all inverse derivatives). *For $0 < v < 1$,*

$$Q^{(n)}(v) = \mathcal{R}_n(Q(v)). \quad (115)$$

Equivalently, for $n \geq 2$,

$$Q^{(n)}(v) = -\frac{1}{F'(Q(v))} \sum_{k=2}^n F^{(k)}(Q(v)) B_{n,k}(Q'(v), \dots, Q^{(n-k+1)}(v)), \quad (116)$$

where $B_{n,k}$ is the partial exponential Bell polynomial.

Proof. The operator relation follows by applying $d/dv = (F'(x))^{-1} d/dx$ at $x = Q(v)$. The Bell recurrence follows by differentiating $F(Q(v)) = v$ n times and isolating the term containing $Q^{(n)}$. $\square$

The first expressions are

$$\begin{aligned} Q' &= \frac{1}{f_1}, \\ Q'' &= -\frac{f_2}{f_1^3}, \\ Q''' &= \frac{3f_2^2 - f_1 f_3}{f_1^5}, \\ Q'''' &= \frac{-15f_2^3 + 10f_1 f_2 f_3 - f_1^2 f_4}{f_1^7}, \end{aligned} \quad (117)$$

with every f_j evaluated at $x = Q(v)$. Combining these formulas with the Lambert–W inverse endpoint expansion from the repository gives asymptotics of any fixed derivative order.

8.3 Derivatives of integral transforms

Since $\mathfrak{M}$ is entire,

$$\mathfrak{M}^{(n)}(s) = \int_0^1 Q(v)^s (\log Q(v))^n dv. \quad (118)$$

The bilateral up transform satisfies, away from its complex zeros,

$$\mathcal{U}^{(n)}(t) = \mathcal{U}(t) B_n(L_1(t), \dots, L_n(t)), \quad (119)$$

where B_n is the complete exponential Bell polynomial and

$$L_j(t) = \frac{d^j}{dt^j} \log \mathcal{U}(t) = \sum_{k=1}^{\infty} 2^{-kj} \ell^{(j)}(t/2^k), \quad \ell(z) = \log \frac{\sinh z}{z}. \quad (120)$$

At $t = 0$, these are precisely the cumulants in [equation \(17\)](#); the continuation of [equation \(119\)](#) through zeros is obtained by cancelling the apparent logarithmic singularities.

9 Endpoint fractional calculus of the inverse Fabius function

The inverse function is the most singular object in the Fabius–Rvachev system. Ordinary derivatives of Q blow up at both endpoints, while Q itself tends to zero more slowly than every positive power. This combination is exactly the regime in which fractional differentiation has a sharp critical exponent. The results in this section are not contained in the audited corpus: the repository develops the Lambert–W inversion and integer inverse derivatives, but not the fractional endpoint law or its strong critical L^p inclusion.

Put

$$T = \log \frac{1}{y}, \quad \ell = \log 2, \quad \eta(y) = \frac{yQ'(y)}{Q(y)}. \quad (121)$$

The dimensionless quantity η is the logarithmic elasticity of the quantile. It is the correct small parameter: powers of $1/T$ alone miss the square-root scale and the repository’s periodic Lambert phase.

Proposition 9.1 (quantile scale and slow variation). *As $y \downarrow 0$,*

$$Q(y) = \frac{\sqrt{T}}{e\sqrt{\ell}} \exp(-\sqrt{2\ell T}) \left(1 + O\left(\frac{\log T}{\sqrt{T}}\right) \right), \quad (122)$$

$$\eta(y) = \sqrt{\frac{\ell}{2T}} + O\left(\frac{\log T}{T}\right). \quad (123)$$

For each fixed $c > 0$,

$$\frac{Q(cy)}{Q(y)} \rightarrow 1. \quad (124)$$

Thus Q , and every positive power Q^β , is slowly varying at the origin in the multiplicative sense.

Proof. Equation (122) is the leading term of the repository’s corrected Lambert–W inverse expansion. The full expansion is differentiable term by term; its logarithmic derivative gives (123). Alternatively, the leading term contributes $\sqrt{\ell/(2T)} - 1/(2T)$ to yQ'/Q , while differentiating the smooth periodic correction contributes $O((\log T)/T)$. Replacing y by cy changes T by the bounded number $-\log c$; the difference of the square roots in (122) therefore tends to zero. This proves (124). $\square$

For $a > 0$, let

$$H_a = \psi(a+1) + \gamma, \quad (125)$$

where ψ is the digamma function and γ is Euler’s constant. Thus $H_n = 1 + 1/2 + \dots + 1/n$ when n is a positive integer.

Theorem 9.2 (digamma correction for fractional quantile primitives). *Let $a > 0$ and $\beta > 0$. Then*

$$I_{0+}^a(Q^\beta)(y) = \frac{y^a Q(y)^\beta}{\Gamma(a+1)} (1 - \beta H_a \eta(y) + o(\eta(y))), \quad y \downarrow 0. \quad (126)$$

Using (123),

$$I_{0+}^a(Q^\beta)(y) = \frac{y^a Q(y)^\beta}{\Gamma(a+1)} \left[1 - \beta H_a \sqrt{\frac{\ell}{2T}} + O\left(\frac{\log T}{T}\right) \right]. \quad (127)$$

For $a = n \in \mathbb{N}$, this gives the first nontrivial endpoint correction to every n -fold antiderivative of every positive quantile power.

Proof. After $v = yu$,

$$I_{0+}^a(Q^\beta)(y) = \frac{y^a Q(y)^\beta}{\Gamma(a)} \int_0^1 (1-u)^{a-1} \left(\frac{Q(yu)}{Q(y)} \right)^\beta du. \quad (128)$$

The differentiated endpoint expansion gives, uniformly on compact logarithmic u -ranges,

$$\log \frac{Q(yu)}{Q(y)} = \eta(y) \log u + o(\eta(y))(1 + |\log u|). \quad (129)$$

Potter bounds for slowly varying functions dominate the remaining small- u tail by an integrable power. We may therefore expand the exponential in (129) under the beta integral. If U_a has density $a(1-u)^{a-1}$ on $(0, 1)$, then

$$\mathbb{E} \log U_a = \psi(1) - \psi(a+1) = -H_a. \quad (130)$$

The constant term in (128) is $1/\Gamma(a+1)$; the linear term is $-\beta H_a \eta(y)$. This proves (126), and (123) gives (127). $\square$

Remark 9.3 (second endpoint correction). Let $\delta(y) = y\eta'(y)$, and put

$$V_a = H_a^2 + \psi_1(1) - \psi_1(a+1), \quad (131)$$

where ψ_1 is the trigamma function. Expanding the logarithmic endpoint jet one order further gives the formal, and under the repository's all-orders smooth expansion rigorous, refinement

$$\frac{\Gamma(a+1) I_{0+}^a(Q^\beta)(y)}{y^a Q(y)^\beta} = 1 - \beta H_a \eta + \frac{V_a}{2} (\beta^2 \eta^2 + \beta \delta) + o(\eta^2 + |\delta|). \quad (132)$$

The appearance of digamma and trigamma values is forced by the first two logarithmic moments of a Beta(1, a) variable. At integer a , these reduce to harmonic and second-order harmonic numbers, linking the fractional expansion directly to the Bernoulli/polygamma organization already present in the repository.

For $0 < \sigma < 1$, define

$$K_\sigma = \psi(1) - \psi(1-\sigma) = -\gamma - \psi(1-\sigma) > 0. \quad (133)$$

Theorem 9.4 (fractional inverse derivative and the sharp critical exponent). *Let $0 < \sigma < 1$ and $\beta > 0$. Since $Q(0) = 0$, the left Caputo and Riemann–Liouville derivatives of Q^β coincide. As $y \downarrow 0$,*

$$D_{0+}^\sigma(Q^\beta)(y) = \frac{y^{-\sigma} Q(y)^\beta}{\Gamma(1-\sigma)} (1 + \beta K_\sigma \eta(y) + o(\eta(y))) \quad (134)$$

$$= \frac{y^{-\sigma} Q(y)^\beta}{\Gamma(1-\sigma)} \left[1 + \beta K_\sigma \sqrt{\frac{\ell}{2T}} + O\left(\frac{\log T}{T}\right) \right]. \quad (135)$$

Consequently, for every $0 < p < \infty$,

$$D_{0+}^{\sigma}(Q^{\beta}) \in L^p(0, \frac{1}{2}) \iff \sigma p \leq 1. \quad (136)$$

By reflection, the same criterion holds for the right derivative $D_{1-}^{\sigma}(1 - Q)^{\beta}$ on $(1/2, 1)$. In particular, the critical exponent $p = 1/\sigma$ is included in the strong L^p space, not merely in weak L^p .

Proof. Apply [theorem 9.2](#) with $a = 1 - \sigma$ and differentiate. Equivalently, compare with the exact model $D_{0+}^{\sigma}y^{\rho} = \Gamma(1 + \rho)y^{\rho-\sigma}/\Gamma(1 + \rho - \sigma)$ and let the local logarithmic exponent ρ tend to $\beta\eta(y)$. The linear coefficient is $\psi(1) - \psi(1 - \sigma) = K_{\sigma}$, which proves (134). The endpoint expansion gives (135).

It remains to decide integrability. The leading asymptotic reduces the question to

$$\int_0^{\varepsilon} y^{-\sigma p} Q(y)^{\beta p} dy. \quad (137)$$

Set $y = e^{-T}$. By (122), the logarithm of the transformed integrand, including $dy = e^{-T} dT$, is

$$(\sigma p - 1)T - \beta p \sqrt{2\ell T} + O(\log T). \quad (138)$$

It tends linearly to $-\infty$ if $\sigma p < 1$, linearly to $+\infty$ if $\sigma p > 1$, and at $\sigma p = 1$ retains the integrable stretched-exponential factor $\exp(-\beta p \sqrt{2\ell T} + O(\log T))$. This proves (136). Reflection follows from $1 - Q(1 - y) = Q(y)$. $\square$

Corollary 9.5 (integer endpoint as a limit). *The limiting case $\sigma \uparrow 1$ predicts and is consistent with*

$$Q' \in L^1(0, 1), \quad Q' \notin L^p(0, 1) \quad (p > 1), \quad (139)$$

which is the known sharp first-inverse-derivative threshold. Thus (136) is a genuine continuous-order interpolation of the integer result.

10 Nonlocal fractional calculus of Rvachev's up-function

Riemann–Liouville operators are one-sided and adapted to antiderivatives. The Riesz potential and fractional Laplacian probe a different aspect: they are translation invariant and nonlocal, so a compactly supported function acquires explicit algebraic tails. For up, those tails are governed by rational moments and therefore by the same Bell–Bernoulli arithmetic as the local theory.

For $0 < s < 1$, use the one-dimensional normalization

$$(-\Delta)^s f(x) = c_{1,s} \text{PV} \int_{\mathbb{R}} \frac{f(x) - f(t)}{|x - t|^{1+2s}} dt, \quad c_{1,s} = \frac{4^s \Gamma(\frac{1}{2} + s)}{\sqrt{\pi} |\Gamma(-s)|}. \quad (140)$$

For $0 < a < 1$, define the Riesz potential

$$(\mathcal{I}_a f)(x) = d_{1,a} \int_{\mathbb{R}} |x - t|^{a-1} f(t) dt, \quad d_{1,a} = \frac{\Gamma(\frac{1-a}{2})}{2^a \sqrt{\pi} \Gamma(\frac{a}{2})}. \quad (141)$$

With the Fourier convention of this report, their multipliers are $|\xi|^{2s}$ and $|\xi|^{-a}$, respectively.

Theorem 10.1 (exact exterior series for the fractional Laplacian). *Let $0 < s < 1$ and $|x| > 1$. Then*

$$(-\Delta)^s \text{up}(x) = -c_{1,s} |x|^{-1-2s} \sum_{j=0}^{\infty} \frac{(1+2s)_{2j}}{(2j)!} \frac{\mu_{2j}}{|x|^{2j}}. \quad (142)$$

The series is absolutely and locally uniformly convergent on $|x| > 1$. Its first terms are

$$(-\Delta)^s \text{up}(x) = -c_{1,s} |x|^{-1-2s} \left[1 + \frac{(1+2s)(1+s)}{9|x|^2} + \frac{19(1+2s)_4}{16200|x|^4} + \dots \right]. \quad (143)$$

Proof. Outside the support, $\text{up}(x) = 0$, so (140) becomes

$$(-\Delta)^s \text{up}(x) = -c_{1,s} \int_{-1}^1 \frac{\text{up}(t)}{|x-t|^{1+2s}} dt. \quad (144)$$

For $x > 1$, expand $(x-t)^{-1-2s} = x^{-1-2s} \sum_{n \geq 0} (1+2s)_n (t/x)^n / n!$. The expansion is uniformly absolutely convergent on every region $x \geq 1 + \varepsilon$. Evenness annihilates all odd moments and gives (142). The case $x < -1$ is the same by symmetry. Substituting $\mu_2 = 1/9$ and $\mu_4 = 19/675$ gives (143). $\square$

Theorem 10.2 (exact exterior Riesz-potential series). *Let $0 < a < 1$ and $|x| > 1$. Then*

$$(\mathcal{I}_a \text{up})(x) = d_{1,a} |x|^{a-1} \sum_{j=0}^{\infty} \frac{(1-a)_{2j}}{(2j)!} \frac{\mu_{2j}}{|x|^{2j}}. \quad (145)$$

Again the series is absolutely and locally uniformly convergent. Thus every exterior coefficient is an explicit gamma factor times a rational Fabius–Rvachev moment.

Proof. For $x > 1$, expand $(x-t)^{a-1} = x^{a-1} (1-t/x)^{-(1-a)}$ in (141); odd moments vanish. Reflection gives $x < -1$. $\square$

Proposition 10.3 (Bell–Bernoulli generation of every coefficient). *The even cumulants of the centered up law are*

$$\kappa_{2r} = \frac{2^{2r} B_{2r}}{2r(2^{2r} - 1)}, \quad \kappa_{2r+1} = 0. \quad (146)$$

Consequently

$$\mu_{2j} = B_{2j}(0, \kappa_2, 0, \kappa_4, \dots, 0, \kappa_{2j}), \quad (147)$$

where B_n is the complete exponential Bell polynomial. Equations (142) and (145) therefore give closed Bernoulli–Bell series for both nonlocal operators.

Proof. Take the logarithm of

$$\mathcal{U}(z) = \prod_{k \geq 1} \frac{\sinh(z/2^k)}{z/2^k}$$

and use

$$\log\left(\frac{\sinh z}{z}\right) = \sum_{r \geq 1} \frac{2^{2r-1} B_{2r}}{r(2r)!} z^{2r}.$$

The geometric factor is $\sum_{k \geq 1} 2^{-2rk} = 1/(2^{2r} - 1)$, which gives (146); the moment–cumulant relation is (147). $\square$

Theorem 10.4 (fractional Sobolev energy as an infinite-sinc integral). *For every $s \geq 0$,*

$$\mathcal{E}_s(\text{up}) := \int_{\mathbb{R}} \text{up}(x) (-\Delta)^s \text{up}(x) dx = \frac{1}{2\pi} \int_{\mathbb{R}} |\xi|^{2s} \prod_{k=1}^{\infty} \text{sinc}^2(\xi/2^k) d\xi. \quad (148)$$

The integral is finite for every s . For $0 < s < 1$, it also equals

$$\mathcal{E}_s(\text{up}) = \frac{c_{1,s}}{2} \iint_{\mathbb{R}^2} \frac{|\text{up}(x) - \text{up}(y)|^2}{|x-y|^{1+2s}} dx dy. \quad (149)$$

Thus the full fractional Sobolev scale of up is encoded by one weighted family of infinite sinc-square integrals.

Proof. Plancherel’s theorem and the Fourier multiplier $|\xi|^{2s}$ give (148). The up-function is C^∞ with compact support, so its Fourier transform decays faster than every power and all energies are finite. The standard symmetrization of the singular integral gives (149). $\square$

11 Fractional self-differential scaling and a dyadic Floquet corrector

The global signed self-differential extension $\mathcal{F}$ satisfies $\mathcal{F}'(x) = 2\mathcal{F}(2x)$. Integer differentiation therefore stays on a single dyadic dilation orbit. The repository's Laplace analysis proves that standard fractional differentiation does not: a nonconstant log-periodic phase is the exact obstruction. Here we retain that obstruction and turn it into a covariant correction.

Let

$$L(s) = \int_0^\infty e^{-sx} \mathcal{F}(x) dx, \quad s > 0. \quad (150)$$

The dilation equation $L(s/2) = sL(s)$ is equivalent to

$$L(2^t) = 2^{-t(t+1)/2} \Pi(t), \quad \Pi(t+1) = \Pi(t), \quad (151)$$

where Π is positive, nonconstant, and has minimal period one. Define the standard Laplace fractional derivative by $\mathcal{L}[\mathbf{D}^\alpha \mathcal{F}](s) = s^\alpha L(s)$ and the continuous dilation orbit by

$$(T_\alpha \mathcal{F})(x) = 2^{\alpha(\alpha+1)/2} \mathcal{F}(2^\alpha x). \quad (152)$$

The known defect identity is

$$\frac{\mathcal{L}[T_\alpha \mathcal{F}](s)}{\mathcal{L}[\mathbf{D}^\alpha \mathcal{F}](s)} = m_\alpha(s), \quad m_\alpha(s) = \frac{\Pi(\log_2 s - \alpha)}{\Pi(\log_2 s)}. \quad (153)$$

In particular, the ratio is identically one exactly for integer α .

Theorem 11.1 (dyadic Floquet corrector and scale cocycle). *Let C_α be the Laplace multiplier with symbol m_α :*

$$\mathcal{L}[C_\alpha f](s) = m_\alpha(s) \mathcal{L}[f](s). \quad (154)$$

Then, on the cyclic vector $\mathcal{F}$,

$$T_\alpha \mathcal{F} = C_\alpha \mathbf{D}^\alpha \mathcal{F} \quad (\alpha \in \mathbb{R}). \quad (155)$$

The symbols satisfy the exact scale-cocycle law

$$m_{\alpha+\beta}(s) = m_\alpha(s/2^\beta) m_\beta(s). \quad (156)$$

Moreover $m_n \equiv 1$ for $n \in \mathbb{Z}$, so the corrector disappears at every integer order.

Proof. Equation (155) is (153) rewritten as an operator identity. For the cocycle,

$$m_\alpha(s/2^\beta) m_\beta(s) = \frac{\Pi(\log_2 s - \beta - \alpha)}{\Pi(\log_2 s - \beta)} \frac{\Pi(\log_2 s - \beta)}{\Pi(\log_2 s)} = m_{\alpha+\beta}(s).$$

Period one gives $m_n = 1$. □

Remark 11.2. The corrected family is not an ordinary translation-invariant semigroup; the argument shift $s \mapsto s/2^\beta$ in (156) is essential. It is better viewed as a Floquet cocycle over logarithmic frequency. This identifies a precise candidate for a fractional calculus adapted to the Fabius dyadic orbit, rather than forcing that orbit into the standard Riemann–Liouville semigroup.

12 Laplace, Fourier, Mellin, and resolvent transforms

12.1 One-sided Laplace transforms

Integration by parts against the distribution measure dF gives, for all $t \in \mathbb{C}$ with removable values at $t = 0$,

$$\int_0^1 e^{tx} F(x) dx = \frac{e^t - P(t)}{t}, \quad (157)$$

$$\int_0^1 e^{tx} (1 - F(x)) dx = \frac{P(t) - 1}{t}. \quad (158)$$

Consequently

$$\int_0^1 x^n e^{tx} F(x) dx = \frac{d^n}{dt^n} \left(\frac{e^t - P(t)}{t} \right), \quad (159)$$

$$\int_0^1 x^n e^{tx} \text{up}(x) dx = \frac{d^n}{dt^n} \left(\frac{P(t) - 1}{t} \right). \quad (160)$$

Purely imaginary t yields the corresponding sine and cosine transforms. On the full line,

$$\int_{-1}^1 e^{tx} \text{up}(x) dx = \mathcal{U}(t), \quad \int_{-1}^1 e^{i\omega x} \text{up}(x) dx = \phi(\omega). \quad (161)$$

12.2 Stieltjes transforms and all resolvent powers

For $z \in \mathbb{C} \setminus [0, 1]$, define

$$G_Y(z) = \mathbb{E} \frac{1}{z - Y} = \int_0^1 \frac{dv}{z - Q(v)}. \quad (162)$$

For $|z| > 1$,

$$G_Y(z) = \sum_{n=0}^{\infty} \frac{m_n}{z^{n+1}}. \quad (163)$$

Similarly, for $z \notin [-1, 1]$,

$$G_Z(z) = \int_{-1}^1 \frac{\text{up}(x)}{z - x} dx = \sum_{n=0}^{\infty} \frac{\mu_n}{z^{n+1}}, \quad |z| > 1. \quad (164)$$

Applying [equation \(77\)](#) to a resolvent primitive gives all higher powers. For every integer $n \geq 1$,

$$\int_0^1 \frac{F(x)}{(z - x)^{n+1}} dx = \frac{1}{n} \left[\frac{1}{(z - 1)^n} - \mathbb{E} \frac{1}{(z - Y)^n} \right], \quad (165)$$

$$\int_0^1 \frac{1 - F(x)}{(z - x)^{n+1}} dx = \frac{1}{n} \left[\mathbb{E} \frac{1}{(z - Y)^n} - \frac{1}{z^n} \right]. \quad (166)$$

For $n = 1$,

$$\int_0^1 \frac{F(x)}{(z - x)^2} dx = \frac{1}{z - 1} - G_Y(z), \quad (167)$$

$$\int_0^1 \frac{1 - F(x)}{(z - x)^2} dx = G_Y(z) - \frac{1}{z}. \quad (168)$$

Using the logarithmic primitive, with a branch analytic off $[0, 1]$,

$$\int_0^1 \frac{F(x)}{z-x} dx = \mathbb{E} \log(z-Y) - \log(z-1), \quad (169)$$

$$\int_0^1 \frac{1-F(x)}{z-x} dx = \log z - \mathbb{E} \log(z-Y). \quad (170)$$

These formulas furnish analytic continuations and boundary-value formulas for Hilbert transforms by approaching the cut from above and below.

12.3 A transform fixed point and the full resolvent hierarchy

The affine random fixed point for the centered up law produces more than a moment series. Let

$$R_n(z) = \int_{-1}^1 \frac{\text{up}(t)}{(z-t)^n} dt, \quad z \in \mathbb{C} \setminus [-1, 1], \quad n \geq 1. \quad (171)$$

Thus $R_1 = G_Z$. If X has density up , then $X \stackrel{d}{=} (X' + U)/2$, where U is uniform on $[-1, 1]$ and independent of $X' \stackrel{d}{=} X$.

Theorem 12.1 (logarithmic fixed point and higher resolvents). *With the logarithm branch normalized at infinity,*

$$R_1(z) = \mathbb{E} \log \left(\frac{2z - X + 1}{2z - X - 1} \right). \quad (172)$$

It follows that

$$R'_1(z) = 2(R_1(2z+1) - R_1(2z-1)), \quad (173)$$

and, for every $n \geq 1$,

$$R_{n+1}(z) = \frac{2^n}{n} (R_n(2z-1) - R_n(2z+1)). \quad (174)$$

For the inverse-Fabius resolvents

$$S_n(z) = \int_0^1 \frac{dv}{(z-Q(v))^n}, \quad z \in \mathbb{C} \setminus [0, 1], \quad (175)$$

we have

$$S_n(z) = 2^n R_n(2z-1), \quad (176)$$

$$S_{n+1}(z) = \frac{2^{n+1}}{n} (S_n(2z-1) - S_n(2z)). \quad (177)$$

Proof. Condition on X' and integrate the uniform variable:

$$\mathbb{E} \left[\frac{1}{z - (X' + U)/2} \middle| X' \right] = \int_{-1}^1 \frac{du}{2z - X' - u} = \log \frac{2z - X' + 1}{2z - X' - 1}.$$

This proves (172). Differentiation gives (173). Since $R_n(z) = (-1)^{n-1} R_1^{(n-1)}(z)/(n-1)!$, differentiating (173) repeatedly gives (174). Finally, $Q(V)$ with $V \sim \text{Unif}[0, 1]$ has the same law as $Y = (X+1)/2$, which gives (176); substitution into (174) gives (177). $\square$

12.4 Entire-kernel calculus

If $h(x) = \sum_{n \geq 0} a_n x^n$ is entire and the series is dominated on the compact support, then

$$\int_{-1}^1 h(x) \text{up}(x) \, dx = \sum_{n=0}^{\infty} a_n \mu_n. \quad (178)$$

$$\int_0^1 h(x) \, dF(x) = \sum_{n=0}^{\infty} a_n m_n. \quad (179)$$

For instance,

$$\int_{-1}^1 J_0(tx) \text{up}(x) \, dx = \sum_{n=0}^{\infty} \frac{(-1)^n (t^2/4)^n}{(n!)^2} \mu_{2n}. \quad (180)$$

Together with [equation \(77\)](#), this gives convergent moment series for Bessel, Airy, hypergeometric, and polylogarithmic kernels without requiring a separate quadrature of the flat endpoint function.

13 Products, order statistics, and correlation integrals

Integrals containing products of F and up are generally not rational, but they have exact probabilistic meanings that lead to useful transform formulas and asymptotics.

13.1 Products of distribution and survival powers

Let $Y_1, \dots, Y_{m+n}$ be independent copies of Y , and define

$$A_m = \max(Y_1, \dots, Y_m), \quad B_n = \min(Y_{m+1}, \dots, Y_{m+n}). \quad (181)$$

The two blocks are independent and

$$\mathbb{P}(A_m \leq x < B_n) = F(x)^m (1 - F(x))^n. \quad (182)$$

Theorem 13.1 (weighted order-statistic gap). *For integers $m, n \geq 1$ and $\Re p > -1$,*

$$\int_0^1 x^p F(x)^m (1 - F(x))^n \, dx = \frac{1}{p+1} \mathbb{E} \left[(B_n^{p+1} - A_m^{p+1}) \mathbf{1}_{\{A_m < B_n\}} \right]. \quad (183)$$

In particular,

$$\int_0^1 F(x)^m (1 - F(x))^n \, dx = \mathbb{E}(B_n - A_m)_+. \quad (184)$$

Proof. Insert the indicator of the event in [equation \(182\)](#), integrate, and apply Fubini. For each realization, the allowed interval is $[A_m, B_n]$ if $A_m < B_n$, and empty otherwise. Its x^p -integral is the right side of [equation \(183\)](#). $\square$

Since $\text{up}(x) = 1 - F(x)$ on $[0, 1]$, this theorem covers all integer products $F^m \text{up}^n$ on the positive half of the support. Two useful special cases are

$$\int_0^1 F(x)^m \, dx = \mathbb{E}(1 - A_m) = \mathbb{E} \min(Y_1, \dots, Y_m), \quad (185)$$

and

$$\int_0^1 F(x)^2 \, dx = \frac{1}{2} - \int_0^1 F(x)(1 - F(x)) \, dx. \quad (186)$$

When the density factor is present, substitution gives a closed beta value for arbitrary complex exponents in their convergence domain:

$$\int_0^1 F(x)^a (1 - F(x))^b F'(x) dx = B(a + 1, b + 1). \quad (187)$$

Using $F'(x) = 2 \operatorname{up}(2x - 1)$,

$$2 \int_0^1 F(x)^a (1 - F(x))^b \operatorname{up}(2x - 1) dx = B(a + 1, b + 1). \quad (188)$$

13.2 Quantiles and ordinary order statistics

If $Y_{r:n}$ is the r -th order statistic of n independent Fabius variables, then $Y_{r:n} = Q(V_{r:n})$ in distribution, where $V_{r:n}$ has beta density. Consequently

$$\mathbb{E}Y_{r:n}^p = \frac{1}{B(r, n - r + 1)} \int_0^1 v^{r-1} (1 - v)^{n-r} Q(v)^p dv. \quad (189)$$

Symmetry gives

$$\mathbb{E}Y_{r:n}^p = \mathbb{E}(1 - Y_{n+1-r:n})^p, \quad \mathbb{E}Y_{r:n} + \mathbb{E}Y_{n+1-r:n} = 1. \quad (190)$$

Expanding the beta polynomial in [equation \(189\)](#) reduces every integer n, r to a finite combination of the mixed quantile integrals appearing in [equation \(23\)](#).

13.3 The Gini integral and the sinc product

For two independent copies Y, Y' , the standard distribution identity is

$$\int_0^1 F(x)(1 - F(x)) dx = \frac{1}{2} \mathbb{E}|Y - Y'|. \quad (191)$$

The difference is $(Z - Z')/2$, and its characteristic function is $\phi(t/2)^2$. The absolute-moment formula

$$\mathbb{E}|X| = \frac{2}{\pi} \int_0^\infty \frac{1 - \Re \varphi_X(t)}{t^2} dt \quad (192)$$

therefore yields an integral containing only the infinite sinc product.

Theorem 13.2 (Gini–sinc identity). *With ϕ as in [equation \(11\)](#),*

$$\boxed{\int_0^1 F(x)(1 - F(x)) dx = \frac{1}{2\pi} \int_0^\infty \frac{1 - \phi(t)^2}{t^2} dt.} \quad (193)$$

Numerically,

$$\int_0^1 F(x)(1 - F(x)) dx = 0.0955632320160066 \dots, \quad (194)$$

so

$$\int_0^1 F(x)^2 dx = 0.4044367679839934 \dots \quad (195)$$

This is a direct link between a nonlinear real-space integral of F and the square of the Rvachev infinite product. It does not appear to simplify to the rational moment constants.

13.4 Autocorrelation and L^2 energy

Define

$$C(a) = \int_{\mathbb{R}} \text{up}(x) \text{up}(x+a) dx. \quad (196)$$

Since up is even, the Wiener–Khinchin formula gives

$$C(a) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-ita} \phi(t)^2 dt. \quad (197)$$

At $a = 0$, Parseval’s identity gives

$$\int_{\mathbb{R}} \text{up}(x)^2 dx = \frac{1}{\pi} \int_0^{\infty} \phi(t)^2 dt = 0.808873535967971 \dots \quad (198)$$

More generally, derivatives of C at the origin are convergent spectral moments whenever the corresponding weighted sinc-product integral converges; the fast decay results developed elsewhere in the repository provide quantitative truncation bounds.

14 Additional integral-transform identities

14.1 A transform relation for inverse resolvents

The general identity [theorem 3.1](#) can be applied in the v -variable as well. Taking $A(x) = x$ and $B(v) = \log(z/(z-v))$ gives

$$\int_0^1 \frac{Q(v)}{z-v} dv + \int_0^1 \log \frac{z}{z-F(x)} dx = \log \frac{z}{z-1}. \quad (199)$$

With $B(v) = (z-v)^{-1}$,

$$\int_0^1 \frac{Q(v)}{(z-v)^2} dv + \int_0^1 \frac{dx}{z-F(x)} = \frac{1}{z-1}. \quad (200)$$

These are nonlinear resolvent dualities: a transform of the quantile is paired with a composition transform of the distribution function.

14.2 Frullani representations for logarithmic constants

For $0 < Y \leq 1$,

$$-\log Y = \int_0^{\infty} \frac{e^{-tY} - e^{-t}}{t} dt. \quad (201)$$

Averaging and using $P(-t)$ gives

$$\int_0^1 \frac{F(x)}{x} dx = \int_0^{\infty} \frac{P(-t) - e^{-t}}{t} dt, \quad (202)$$

where

$$P(-t) = \prod_{k=1}^{\infty} \frac{1 - e^{-t/2^k}}{t/2^k}. \quad (203)$$

For $a > 0$, the negative moment itself has the Mellin–Laplace representation

$$\mathfrak{M}(-a) = \frac{1}{\Gamma(a)} \int_0^{\infty} t^{a-1} P(-t) dt. \quad (204)$$

This is an independent numerical route to [equation \(44\)](#) and connects the negative-moment constant C_* to the dyadic product on the positive Laplace axis.

14.3 A smoothing difference equation for $\mathfrak{M}$

Let

$$N(s) = \mathbb{E}(1 + Y)^s = \sum_{k=0}^{\infty} \binom{s}{k} m_k. \quad (205)$$

The series is entire because of [theorem 4.3](#). Integrating the uniform variable in [equation \(6\)](#) with an arbitrary complex exponent gives the entire difference relation

$$(s + 1)2^s \mathfrak{M}(s) = N(s + 1) - \mathfrak{M}(s + 1), \quad (206)$$

with the value at $s = -1$ interpreted by continuity. At nonnegative integer s , coefficient comparison recovers [equation \(14\)](#); at negative s , it is equivalent to [equation \(55\)](#). Thus the positive rational moments and the superexponentially growing negative moments are two sides of one entire difference equation.

15 Numerical experiments and reproducibility

The accompanying program `fractional_frontier_experiments.py` is fully commented and deterministic. It reconstructs up by FFT inversion of the sinc product on a period-four grid with 2^{18} points; the support $[-1, 1]$ prevents periodic overlap. The Fabius function is then obtained from $F(x) = \text{up}(x - 1)$ on $[0, 1]$, and a monotone inverse gives Q . No Monte Carlo sampling or network access is used.

The program independently checks four parts of the new analysis:

1. the exterior series [\(142\)](#) and [\(145\)](#) against direct quadrature of their defining kernels;
2. the logarithmic Cauchy fixed point [\(172\)](#), differential equation [\(173\)](#), and hierarchy [\(174\)](#);
3. the normalized endpoint ratio in [\(126\)](#) and its first digamma correction;
4. the normalized fractional derivative in [\(134\)](#). The derivative is obtained by a symmetric five-point differentiation of $I^{1-\sigma}Q$, avoiding the severe conditioning loss in direct division by the flat density F' .

s	$ x $	direct $(-\Delta)^s \text{up}$	moment series	absolute error
0.25	1.20	$-1.80996700e - 01$	$-1.80996689e - 01$	$1.07e - 08$
0.25	1.50	$-1.20466958e - 01$	$-1.20466958e - 01$	$1.12e - 12$
0.25	2.00	$-7.45381034e - 02$	$-7.45381034e - 02$	$5.55e - 17$
0.50	1.20	$-2.95515474e - 01$	$-2.95515385e - 01$	$8.94e - 08$
0.50	1.50	$-1.67488207e - 01$	$-1.67488207e - 01$	$8.18e - 12$
0.50	2.00	$-8.70075045e - 02$	$-8.70075045e - 02$	$8.33e - 17$
0.75	1.20	$-2.93003324e - 01$	$-2.93002949e - 01$	$3.75e - 07$
0.75	1.50	$-1.39364026e - 01$	$-1.39364026e - 01$	$3.02e - 11$
0.75	2.00	$-6.03002358e - 02$	$-6.03002358e - 02$	$2.15e - 16$

Table 2: Exterior fractional-Laplacian series checked against direct FFT-grid quadrature.

At $|x| = 1.2$, where the exterior moment series converges most slowly among the displayed points, eighteen exact moments already give errors between 10^{-8} and 4×10^{-7} for the fractional Laplacian; at $|x| = 2$, the discrepancy is at floating-point roundoff. The Cauchy-transform identities have residuals below 10^{-16} . In the endpoint tables, the first digamma correction reduces the primitive error from about 10^{-1} to about 2×10^{-2} , and the fractional-derivative error from about 1.4×10^{-1} to about 1.5×10^{-2} at the smallest resolved y . The trend is consistent with the predicted $O((\log T)/T)$ remainder; FFT resolution, rather than the asymptotic formula, becomes the limiting factor farther into the flat endpoint.

α	$ x $	direct $\mathcal{I}_\alpha$ up	moment series	absolute error
0.25	1.20	$1.38089169e-01$	$1.38089169e-01$	$4.12e-10$
0.25	1.50	$1.14103403e-01$	$1.14103403e-01$	$5.18e-14$
0.25	2.00	$9.04704471e-02$	$9.04704471e-02$	$4.16e-17$
0.50	1.20	$3.76439711e-01$	$3.76439711e-01$	$3.17e-10$
0.50	1.50	$3.32338157e-01$	$3.32338157e-01$	$4.24e-14$
0.50	2.00	$2.85179706e-01$	$2.85179706e-01$	$1.11e-16$
0.75	1.20	$1.03262489e+00$	$1.03262489e+00$	$1.72e-10$
0.75	1.50	$9.71451305e-01$	$9.71451305e-01$	$2.50e-14$
0.75	2.00	$9.00632222e-01$	$9.00632222e-01$	$1.11e-16$

Table 3: Exterior Riesz-potential series checked against direct quadrature.

y	$Q(y)$	$\eta(y)$	normalized integral	first correction	corrected error
$1.0e-02$	$1.55942014e-01$	$2.18739e-01$	0.87083647	0.83696799	$3.39e-02$
$3.0e-03$	$1.21418852e-01$	$1.97703e-01$	0.88052201	0.85264653	$2.79e-02$
$1.0e-03$	$9.85392757e-02$	$1.83167e-01$	0.88730767	0.86348039	$2.38e-02$
$3.0e-04$	$7.96341581e-02$	$1.71106e-01$	0.89338014	0.87246987	$2.09e-02$
$1.0e-04$	$6.63361227e-02$	$1.61734e-01$	0.89815279	0.87945530	$1.87e-02$

Table 4: Fractional quantile primitive with $\alpha = 0.65$ and $\beta = 1.0$. The normalized integral is $\Gamma(\alpha + 1)I^\alpha(Q^\beta)/(y^\alpha Q(y)^\beta)$.

y	$Q(y)$	$\eta(y)$	normalized derivative	first correction	corrected error
$1.0e-02$	$1.55942014e-01$	$2.18739e-01$	1.18349375	1.21073362	$2.72e-02$
$3.0e-03$	$1.21418852e-01$	$1.97703e-01$	1.16810670	1.19046769	$2.24e-02$
$1.0e-03$	$9.85392757e-02$	$1.83167e-01$	1.15744377	1.17646395	$1.90e-02$
$3.0e-04$	$7.96341581e-02$	$1.71106e-01$	1.14819379	1.16484423	$1.67e-02$
$1.0e-04$	$6.63361227e-02$	$1.61734e-01$	1.14094770	1.15581494	$1.49e-02$

Table 5: Caputo/Riemann–Liouville quantile derivative with $\sigma = 0.4$ and $\beta = 1.0$. The normalized derivative is $\Gamma(1 - \sigma)y^\sigma D^\sigma(Q^\beta)/Q(y)^\beta$.

transform identity	absolute residual
logarithmic Cauchy fixed point	$0.000e+00$
Cauchy differential equation	$7.850e-17$
resolvent hierarchy, $m = 1$	$7.850e-17$
resolvent hierarchy, $m = 2$	$2.776e-17$
resolvent hierarchy, $m = 3$	$3.925e-17$

Table 6: Residuals for the transform fixed-point and higher-resolvent identities at $z = 1.35 + 0.60i$.

The script writes a human-readable ledger `numerical_results.txt`, a JSON file `numerical_results.json`, and the LaTeX tables included above. Running

```
python fractional_frontier_experiments.py
```

regenerates all three files.

16 Conjectures and frontier directions

16.1 Critical fractional Sobolev regularity of the inverse

Conjecture 16.1 (sharp fractional Sobolev boundary). *For $0 < s < 1$, $1 \leq p < \infty$, and $\beta > 0$,*

$$Q^\beta \in W^{s,p}(0,1) \iff sp \leq 1. \quad (207)$$

At the critical line $sp = 1$, membership should be strong rather than merely weak, with the stretched-exponential inverse endpoint factor providing the additional integrability. The one-sided derivative theorem [theorem 9.4](#) proves the corresponding statement in the Riemann–Liouville endpoint scale. A proof of (207) should combine a fractional Hardy equivalence with a two-sided localization and the repository’s all-orders inverse expansion.

16.2 Function-space realization of the Floquet corrector

The symbol $m_\alpha(s)$ in (153) is bounded above and below on the positive axis for each fixed real α , because Π is positive and periodic. It is therefore natural to expect a bounded causal operator on weighted Laplace–Hardy or Mellin–Sobolev spaces.

Conjecture 16.2 (dyadic fractional calculus). *There is a natural Banach scale $\mathcal{X}_\rho$ containing the global signed Fabius extension such that:*

1. *every C_α extends boundedly and invertibly to $\mathcal{X}_\rho$;*
2. *the scale-cocycle identity (156) becomes an operator cocycle under dyadic dilation;*
3. *$C_\alpha \mathbf{D}^\alpha$ is the unique continuous-order extension of the integer derivative orbit satisfying the cocycle and a mild spectral boundedness condition.*

The Mellin variable $\log_2 s$ turns the problem into periodic Floquet theory, while the nonzero Fourier modes of $\log \Pi$ should control norm distortion.

16.3 Arithmetic nonlocal tails

For rational s or a , the series in [theorems 10.1](#) and [10.2](#) has rational moment coefficients multiplied by explicit gamma constants. The arithmetic nature of special values such as $(-\Delta)^{1/2} \text{up}(2)$ and $(\mathcal{I}_{1/2} \text{up})(2)$ is unknown.

Conjecture 16.3 (noncollapse of exterior nonlocal values). *Except for identities forced by gamma reflection and the rational moment recurrence, the special values of the exterior fractional-Laplacian and Riesz-potential series at rational $|x| > 1$ do not reduce to finite combinations of algebraic numbers and powers of π . A first tractable target is linear independence of several values over the field generated by the relevant gamma constants.*

16.4 Near-rigidity of the negative-moment phase

Conjecture 16.4 (phase rigidity). *The continuous one-periodic limiting profile $C(\theta)$ in [equation \(62\)](#) is constant:*

$$C(\theta) \equiv C_*. \quad (208)$$

A weaker and perhaps more realistic alternative is that C is nonconstant but its first Fourier harmonics are suppressed by the same spectral mechanism responsible for the extremely small periodic oscillations in the endpoint asymptotics studied elsewhere in the repository. The exact recurrence reduces this question to the Fourier analysis of the periodic product [equation \(62\)](#). A rigorous high-precision test should compute $A(\theta)$ independently by [equation \(204\)](#), rather than by the same moment data used in the recurrence.

16.5 Completely regular growth of the entire Mellin function

The maximum modulus is known exactly through [equation \(72\)](#), but directional cancellation remains open.

Conjecture 16.5 (indicator function). *The entire function $\mathfrak{M}$ has completely regular growth of order 2 with indicator*

$$h(\vartheta) = \frac{\log 2}{2} \max\{0, -\cos \vartheta\}^2. \quad (209)$$

The upper bound follows immediately from $|\mathfrak{M}(re^{i\vartheta})| \leq \mathfrak{M}(r \max(0, -\cos \vartheta))$. The difficult part is a matching lower bound away from the negative real axis, where oscillation in $e^{i\Im(s) \log Y}$ may create zeros and exceptional directions.

16.6 Arithmetic nature of the new constants

The quantities

$$C_*, \quad \mathfrak{M}(-r), \quad -\mathbb{E} \log Y, \quad \int_0^1 F(1-F), \quad \int_{\mathbb{R}} \text{up}^2 \quad (210)$$

are defined by positive rational series or infinite sinc products, but no reduction to standard constants is visible.

Conjecture 16.6. *The constants C_* , $\mathfrak{M}(-r)$ for every positive integer r , and $-\mathbb{E} \log Y$ are irrational; at least one of the stronger properties of transcendence or algebraic independence should hold among them.*

This is deliberately stated as a research target, not as numerical evidence for a specific transcendence theorem. Possible tools include Mahler-type functional equations for $P(t)$, arithmetic estimates for the rational Newton partial sums, and Padé approximants controlled by the moment recurrence.

16.7 Complete monotonicity of normalized errors

The sequence C_r is decreasing by [equation \(66\)](#). Numerics suggest much more.

Conjecture 16.7. *The error sequence $C_r - C_*$ is completely monotone:*

$$(-1)^k \Delta^k (C_r - C_*) \geq 0 \quad (r \geq 1, k \geq 0). \quad (211)$$

A proof might represent $C_r - C_*$ as a Hausdorff moment sequence derived from the positive tail product $\prod_{j \geq r} (1 - \rho(j))$.

16.8 Order-statistic endpoint regimes

When the sample size n grows, the minimum $Y_{1:n}$ probes precisely the Lambert– W endpoint regime. Since

$$\mathbb{P}(Y_{1:n} > x) = (1 - F(x))^n,$$

its moments are

$$\mathbb{E}Y_{1:n}^p = p \int_0^1 x^{p-1} (1 - F(x))^n dx. \quad (212)$$

A promising direction is a uniform saddle-point expansion in p and n , with the periodic endpoint phase transported into order-statistic oscillations. The corresponding maximum follows by symmetry. This would connect the present integral calculus directly to the repository's inverse-Lambert and half-integer spectral programs.

16.9 General base and q -deformations

Replace the dyadic weights by b^{-k} , normalized so the support remains $[0, 1]$. The distributional smoothing, inverse-pair calculus, and fractional identities survive unchanged, while the product $P(t)$, Bernoulli cumulants, and negative-moment normalization acquire base b . The expected leading law is

$$\mathbb{E}Y_b^{-r} \asymp C_b \frac{b^{r(r+1)/2}}{\Gamma(r)}. \quad (213)$$

Determining C_b , its phase profile, and its relationship with $q = 1/b$ binomial coefficients is a natural extension.

16.10 Formalization roadmap

The most accessible Lean targets are:

1. the measure-theoretic quantile transport and [theorem 3.1](#);
2. the power identity [equation \(23\)](#) and its polynomial corollaries;
3. the positive and negative moment recurrences;
4. positivity, monotonicity, and convergence of C_r ;
5. the order-statistic gap identity;
6. finite truncations and certified tail bounds for the Newton and sinc-product series.

The entire-function and fractional-integral portions require more analytic infrastructure, but can be decomposed into dominated-convergence, holomorphic-parameter integration, and beta-integral lemmas.

17 Summary of principal formulas

For convenient reference, the central formulas are collected here.

Distribution and transforms

$$Y = \sum_{k \geq 1} 2^{-k} U_k, \quad Z = 2Y - 1,$$

$$P(t) = \prod_{k \geq 1} \frac{e^{t/2^k} - 1}{t/2^k}, \quad \phi(t) = \prod_{k \geq 1} \frac{\sin(t/2^k)}{t/2^k},$$

$$\int_0^1 e^{tx} F(x) dx = \frac{e^t - P(t)}{t}, \quad \int_0^1 e^{tx} (1 - F(x)) dx = \frac{P(t) - 1}{t}.$$

Inverse-pair integration

$$A(z)B(F(z)) - A(0)B(0) = \int_0^z A' B \circ F + \int_0^{F(z)} A \circ Q B',$$

$$p \int_0^1 x^{p-1} F(x)^q dx + q \int_0^1 v^{q-1} Q(v)^p dv = 1,$$

$$s \int_0^1 e^{sx+tF(x)} dx + t \int_0^1 e^{tv+sQ(v)} dv = e^{s+t} - 1.$$

Entire moment calculus

$$\mathfrak{M}(s) = \int_0^1 Q(v)^s dv,$$

$$\mathfrak{M}(s) = \sum_{k \geq 0} (-1)^k \binom{s}{k} m_k = 2^{-s} \sum_{j \geq 0} \binom{s}{2j} \mu_{2j},$$

$$\int_0^1 x^{s-1} F(x) dx = \frac{1 - \mathfrak{M}(s)}{s}, \quad \int_0^1 x^{s-1} \text{up}(x) dx = \frac{\mathfrak{M}(s)}{s}.$$

Negative moments and fractional primitives

$$A(s+1) = \frac{2^{s+1}}{s} (A(s) - D(s)),$$

$$\mathfrak{M}(-r) \sim C_* \frac{2^{r(r+1)/2}}{\Gamma(r)},$$

$$I_0^\alpha F^q(x) = \frac{q}{\Gamma(\alpha+1)} \int_0^{F(x)} (x - Q(v))^\alpha v^{q-1} dv,$$

$$I_0^\alpha Q^p(v) = \frac{p}{\Gamma(\alpha+1)} \int_0^{Q(v)} (v - F(x))^\alpha x^{p-1} dx.$$

New fractional endpoint laws

For $T = \log(1/y)$, $\eta = yQ'/Q \sim \sqrt{\log 2/(2T)}$,

$$I_{0+}^a (Q^\beta)(y) = \frac{y^a Q(y)^\beta}{\Gamma(a+1)} (1 - \beta H_a \eta + o(\eta)),$$

$$D_{0+}^\sigma (Q^\beta)(y) = \frac{y^{-\sigma} Q(y)^\beta}{\Gamma(1-\sigma)} (1 + \beta K_\sigma \eta + o(\eta)),$$

$$D_{0+}^\sigma (Q^\beta) \in L^p(0, \frac{1}{2}) \iff \sigma p \leq 1.$$

New nonlocal and resolvent formulas

For $|x| > 1$,

$$(-\Delta)^s \text{up}(x) = -c_{1,s} |x|^{-1-2s} \sum_{j \geq 0} \frac{(1+2s)_{2j}}{(2j)!} \frac{\mu_{2j}}{|x|^{2j}},$$

$$(\mathcal{I}_a \text{up})(x) = d_{1,a} |x|^{a-1} \sum_{j \geq 0} \frac{(1-a)_{2j}}{(2j)!} \frac{\mu_{2j}}{|x|^{2j}},$$

$$R_{n+1}(z) = \frac{2^n}{n} (R_n(2z-1) - R_n(2z+1)).$$

The fractional self-differential orbit is restored exactly by the log-periodic multiplier $m_\alpha(s) = \Pi(\log_2 s - \alpha) / \Pi(\log_2 s)$, which obeys a dyadic scale-cocycle law.

18 Conclusion

The probability/quantile viewpoint converts the integral theory of the Fabius–Rvachev system into a coherent calculus rather than a list of isolated evaluations. The inverse-pair identity organizes indefinite weighted integrals; the entire moment function organizes fractional and negative powers; Riemann–Liouville operators organize repeated antiderivatives; and the infinite sinc product organizes correlation and fractional Sobolev energies. The inverse endpoint has a particularly clean continuous-order law: digamma coefficients interpolate the harmonic-number corrections, and the line $\sigma p = 1$ is the exact strong integrability boundary. Nonlocal differentiation creates algebraic tails whose complete coefficient sequence remains rational before the universal gamma normalization. The negative-moment recurrence is also especially useful: it extracts an additional constant and an analytic periodic phase directly from the dyadic smoothing law, with exponentially fast normalization.

The next decisive steps are to determine whether the phase profile is exactly constant, to connect its Fourier modes with the spectral corrections already present in endpoint asymptotics, and to formalize the inverse-pair and moment recurrences. These directions are sufficiently concrete to support both high-precision experiments and Lean proofs.

A Exact low-order data

The exact positive moments used by the script are

n	$m_n = \mathbb{E}Y^n$	$\mu_n = \mathbb{E}Z^n$ when shown
0	1	1
1	1/2	0
2	5/18	1/9
3	1/6	0
4	143/1350	19/675
5	19/270	0
6	1153/23814	583/59535
7	583/17010	0
8	1616353/65063250	132809/32531625
9	132809/7229250	0
10	134926369/9762090030	46840699/24405225075

The centered entries can be generated more economically from $\mu_{2j} = 2m_{2j+1}/(2j + 1)$ than by expanding $(2Y - 1)^{2j}$, which avoids severe cancellation at high order.

B Repository audit categories

The recursive review used the following categories.

Category	Role in the audit
Core expositions	Definitions, probability law, smoothness, dyadic derivatives, symmetry, Fourier product, moment arithmetic.
Glossary and Lean walkthrough	Normalization checks and correspondence with formalized objects.
Canonical research-frontier volume	Primary non-duplication comparison for asymptotics, Lambert–W, Bell/Bernoulli, q -binomial, interpolation, and extrapolation themes.
Current drafts	Checks against dyadic self-sampling, inverse-function expansions, Rvachev–Appell systems, half-integer spectral effects, Fourier decay, and certification algorithms.
Papers	Historical and arithmetic results, exact dyadic values, Fourier coefficients, and endpoint asymptotics.
Frozen archive	Historical variants and absorbed integration material, checked through the archive map and canonical successors to prevent counting old snapshots as independent frontiers.

C Recursive TeX audit ledger

The following ledger records the authoritative sixty-seven-path pinned inventory, the eleven-path live-main supplement, the resulting seventy-eight-path current count, and the integration-specific exclusion searches used to avoid duplication.

Recursive TeX audit ledger Repository: VladimirReshetnikov/ProveIt Root: Analysis/FabiusFunction/docs Audit date: 2026-08-27 Scope and counting convention ----- The audit has two layers. (1) A path-complete pinned baseline at commit 32d6d36c51d803289e6d6a0dc0c37753766eba47 contains exactly 67 distinct *.tex paths. Its authoritative inventory is Appendix A of Exponent_Sequence_Newton_Rvachev_Frontiers.tex. (2) The live main-branch tree was then traversed for later additions. Five later draft packages add 11 visible *.tex paths: seven files in the dyadic self-sampling package and one file each in the inverse, Newton/exponent, dyadic-inverse/Barnes, and q-Richardson/sinc packages. Thus the current tree inspected for this report contains 78 visible *.tex paths: 5 core synthesis/reference sources 32 frozen historical sources 34 active frontier-report or auxiliary TeX sources 7 paper transcriptions -- 78 total Generated table fragments were read together with their parent reports and underlying data. They were counted as TeX paths but not as independent mathematical claims. Frozen revisions were read for distinctive claims and provenance, but repeated theorems were counted once when setting the novelty boundary. A. Authoritative 67-path pinned baseline ----- 1. FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex 2. Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
--

3. Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4. fabius_lean_walkthrough/fabius_lean_walkthrough.tex
5. non-formalized-research-frontiers/non-formalized-research-frontiers.tex
6. archive/early-notes/Fabius_Asymptotic/Fabius_Asymptotic.tex
7. archive/early-notes/K-fold summation over the signed Thue-Morse sequence/K-fold summation over the signed Thue-Morse sequence.tex
8. archive/glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex
9. archive/glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
10. archive/lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex
11. archive/lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
12. archive/primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex
13. archive/primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex
14. archive/primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
15. archive/primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex
16. archive/primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex
17. archive/primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex
18. archive/primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex
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25. archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex
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27. archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex
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37. archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex
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43. non-formalized-research-frontiers/drafts/Fabius_Rvachev_Thue_Morse_Frontier_Results/Fabius_Rvachev_Thue_Morse_Frontier_Results.tex
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49. non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex
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61. papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
62. papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
63. papers/arXiv-1502.06738v2/Lacunary_products.tex
64. papers/arXiv-1702.05442v1/09-Function.tex
65. papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
66. papers/arXiv-2211.13570v1/document.tex
67. papers/ubiq15/ubiq15-corrected.tex

B. Live-main supplement after the pinned baseline (11)

68. non-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/Fabius_Dyadic_Self_Sampling_Frontier_Report.tex
69. non-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/appell_polynomials.tex
70. non-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/harmonic_tail_table.tex
71. non-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/quadrature_table.tex
72. non-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/quadrature_table_display.tex
73. non-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/spectral_check.tex
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78. non-formalized-research-frontiers/drafts/fabius_frontier_results/q_richardson_sinc_frontier.tex

Integration-specific comparison protocol

The primary exposition, canonical frontier synthesis, all active report sources, auxiliary TeX fragments, paper transcriptions, and frozen integration/inverse notebooks were compared at theorem and section level. Targeted literal and close-variant searches included:

antiderivative; indefinite integral; repeated primitive; weighted primitive;
inverse-pair area identity; quantile integral; order statistic; beta transform;
Mellin transform; Stieltjes transform; Cauchy transform; resolvent hierarchy;
negative moment; complex moment; Newton series; central moment series;
Riemann--Liouville; Caputo; fractional integral; fractional derivative;

fractional Laplacian; Riesz potential; Sobolev energy; Gagliardo seminorm;
digamma; trigamma; critical L_p ; Floquet multiplier; dilation cocycle;
Gini mean difference; autocorrelation; sinc-square integral.

Prior-art boundary

The audit treats as prior input: the probability/refinement definitions; exact dyadic arithmetic; q -binomial and Thue--Morse identities; rational moment and Bernoulli/Bell systems; the infinite sinc product and its spectral/Fourier-decay analysis; repeated integration and Rvachev--Appell systems; Lambert-W endpoint and inverse asymptotics; Newton/exponent-sequence and q -Richardson machinery; self-sampling quadrature; and the elementary weighted/inverse integration identities already absorbed into the canonical frontier volume.

The strongest corpus-relative frontier claims of the accompanying report are therefore restricted to the explicitly proved fractional endpoint corrections, sharp critical L_p law, nonlocal exterior series and fractional energy identity, Cauchy/logarithmic transform fixed point and resolvent hierarchy, and the Floquet multiplier/cocycle that repairs fractional dyadic scaling. This is a repository-relative nonduplication statement, not a claim of worldwide historical priority.

D Reproducibility instructions

From the directory containing this report, run

```
python fractional_frontier_experiments.py
```

to regenerate `new_numerical_results.tex`, `numerical_results.txt`, and `numerical_results.json`. Rebuild the PDF with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
    fabius_fractional_integral_transform_frontiers.tex
```

The experiment uses NumPy, SciPy, and mpmath. All exact moments are represented by `Fraction`; the FFT and quadratures are used only as independent floating-point checks of proved identities.

References

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